

AP Precalculus 2026–2027

Unit 3: Trigonometric and Polar Functions

Lesson 3.8: The Tangent Function

(2 instructional periods, 55 minutes each)

Primary Objective

Big Idea: Functions

Students construct representations of the tangent function using the unit circle, describe its key characteristics (period π , vertical asymptotes, monotone increase, concavity change), and analyze the effect of additive and multiplicative transformations on the tangent graph.

Priority Ideas & Skills

Skills: 2.A (Identify information from representations) 3.A (Describe characteristics of functions)

Essential Knowledge:

- **EK 3.8.A.1/A.2** $f(\theta) = \tan \theta$ gives the slope of the terminal ray; $\tan \theta = \sin \theta / \cos \theta$ where $\cos \theta \neq 0$.
- **EK 3.8.B.1** Period is π — slope of the terminal ray repeats every half-revolution.
- **EK 3.8.B.2** Vertical asymptotes at $\theta = \pi/2 + k\pi$, $k \in \mathbb{Z}$, because $\cos \theta = 0$ there.
- **EK 3.8.B.3** The tangent function is always increasing between consecutive asymptotes; its graph changes from concave down to concave up within each period.
- **EK 3.8.C.1–C.5** Transformations $g(\theta) = a \cdot \tan(b(\theta + c)) + d$: vertical dilation $|a|$, period $\pi/|b|$, phase shift $-c$, vertical translation d .

Vocabulary, Concepts, & Theorems

- | | | |
|-------------------------|---------------------|--------------------|
| • tangent function | • period (tangent) | • concavity |
| • slope of terminal ray | • phase shift | • inflection point |
| • vertical asymptote | • vertical dilation | |

Activate Prior Knowledge & Spiral Review

- **Lessons 3.2–3.4** — Unit circle: $\sin \theta$ and $\cos \theta$ values at multiples of $\pi/6$ and $\pi/4$; interpreting a point on the unit circle as $(\cos \theta, \sin \theta)$.
- **Algebra** — Slope of a line through the origin: rise over run.
- **Pre-3.8** — Identifying vertical asymptotes of rational functions; behavior near a denominator zero.

Hook

“A lighthouse beam sweeps the shore. A sensor at the end of the dock records how far along the shore the beam strikes as a function of the angle θ the beam makes with the perpendicular. When the beam is nearly parallel to the shore, the illuminated point races off to infinity. What function models this?”

- If the dock is 1 unit from the lighthouse, the distance along the shore is $\tan \theta$.
- As $\theta \rightarrow \pi/2$, $\tan \theta \rightarrow +\infty$: the beam sweeps to infinity — an asymptote in action.
- **Key idea:** the tangent function is the slope of the terminal ray of angle θ on the unit circle.

Transition: Today we derive $\tan \theta = \sin \theta / \cos \theta$ from the unit circle, describe all key characteristics of the tangent graph, and explore how transformations shift and stretch it.

Lesson — Period 1: Deriving and Graphing Tangent

Tangent as Slope of Terminal Ray

Place angle θ in standard position. The terminal ray passes through $(\cos \theta, \sin \theta)$ and the origin. Its slope is:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Defined whenever $\cos \theta \neq 0$.

Key Values Table

θ	$\tan \theta$
0	0
$\pi/4$	1
$\pi/3$	$\sqrt{3}$
$\pi/2$	undefined
$2\pi/3$	$-\sqrt{3}$
$3\pi/4$	-1
π	0

Why period = π : After a half-revolution the terminal ray points in the opposite direction, but its slope is identical: the rise and run both change sign, so the ratio is unchanged. Thus $\tan(\theta + \pi) = \tan \theta$.

Asymptotes: When $\cos \theta = 0$ (i.e. $\theta = \pi/2 + k\pi$), the slope is undefined and $|\tan \theta|$ grows without bound — vertical asymptotes at each of these values.

Lesson (cont.) — Period 2: Characteristics & Transformations

Characteristics in $(-\pi/2, \pi/2)$

- Passes through $(0, 0)$; always increasing.
- Concave down for $\theta < 0$; inflection at $(0, 0)$; concave up for $\theta > 0$.
- $\tan \theta \rightarrow +\infty$ as $\theta \rightarrow (\pi/2)^-$.
- $\tan \theta \rightarrow -\infty$ as $\theta \rightarrow (-\pi/2)^+$.

Transformations: $g(\theta) = a \tan(b(\theta + c)) + d$

Parameter	Effect
a	vertical dilation by $ a $
b	period becomes $\pi/ b $
c	phase shift of $-c$
d	vertical translation by d

Finding new asymptotes: Set $b(\theta + c) = \pi/2 + k\pi$ and solve for θ . Example: for $g(\theta) = \tan(2\theta)$, asymptotes where $2\theta = \pi/2 + k\pi$, so $\theta = \pi/4 + k\pi/2$.

Explicit Instruction: Describing Tangent Transformations

Steps:

1. Identify a, b, c, d from $g(\theta) = a \tan(b(\theta + c)) + d$.
2. State period: $\pi/|b|$.
3. Find asymptotes: solve $b(\theta + c) = \pm\pi/2$.
4. State phase shift $-c$ and vertical shift d .
5. State vertical dilation: output values scaled by $|a|$.

Worked Example: $g(\theta) = 2 \tan(\frac{1}{2}(\theta - \frac{\pi}{4})) + 1$

1. $a = 2, b = 1/2, c = -\pi/4, d = 1$.
2. Period = $\pi \div (1/2) = 2\pi$.
3. Asymptotes: $\frac{1}{2}(\theta - \frac{\pi}{4}) = \pm\frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4} \pm \pi$, so $\theta = \frac{5\pi}{4}$ and $\theta = -\frac{3\pi}{4}$ (plus integer multiples of 2π).
4. Phase shift: $+\pi/4$ (right); vertical shift: up 1.
5. All output values doubled (vertical dilation by 2).

Active Monitoring

Circulate as students work. Watch for:

- Confusing the period of tangent (π) with that of sine/cosine (2π).
- Placing asymptotes at multiples of π instead of $\pi/2 + k\pi$.
- Treating $|a|$ as an “amplitude” (tangent has no amplitude).
- Forgetting to divide by $|b|$ when computing the new period.

Cold-call: “What is the period of $\tan(3\theta)$? Where are its asymptotes?” (Period = $\pi/3$; asymptotes at $\theta = \pi/6 + k\pi/3$.)

Group Work & Differentiation

Students work in assigned tiers on the Group Activity.

- **Tier R (Remediate):** Given unit-circle $\sin / \cos$ values, compute $\tan \theta$ at key angles and identify asymptote locations.
- **Tier A (Approaching Proficiency):** Given a tangent graph, identify period, asymptotes, and concavity; describe the transformation $g(\theta) = a \tan(\theta) + d$.
- **Tier E (Extension):** Analyze $g(\theta) = 2 \tan(\theta/2 - \pi/4) + 1$: identify all parameters, describe the full transformation, find asymptotes, and evaluate g at two specific angles.

Individual Work & Assessment

- **Exit Ticket** (last 5–7 min): 3 short items — computing $\tan \theta$ from given $\sin / \cos$ values, identifying the period of a transformed tangent, and locating one asymptote.
- **Homework:** 5–6 problems including an AP-style MC and a preview of Lesson 3.9 (inverse trigonometric functions).
- **Formative data:** Collect exit tickets; students who misplace asymptotes need reteaching before Lesson 3.9.

Reinforcement & Extension

Reinforce: Return to the unit circle. Draw several terminal rays and compute their slopes directly. The slope of the ray through the origin and $(\cos \theta, \sin \theta)$ is $\sin \theta / \cos \theta$ — tangent is slope.

Extension: Use Desmos to graph $y = a \tan(\theta)$ for several values of a . Ask: as $|a|$ increases, how does the graph steepen near its inflection point?

Preview — Lesson 3.9: Inverse Trigonometric Functions

In Lesson 3.9 students restrict the domains of sine, cosine, and tangent to make them invertible, define $\arcsin$, $\arccos$, and $\arctan$, and evaluate inverse trig expressions — building directly on today’s asymptote behavior of $\tan \theta$ near $\pm\pi/2$.